ECE 2305 – Introduction to C Programming
Programming Project 02
Quadratic Equation Solver

Program Features: Branch structure, variables, data input and output, mathematical operators, function calls.

Design a C++ application that calculates and displays all solutions to quadratic algebraic equations of the form

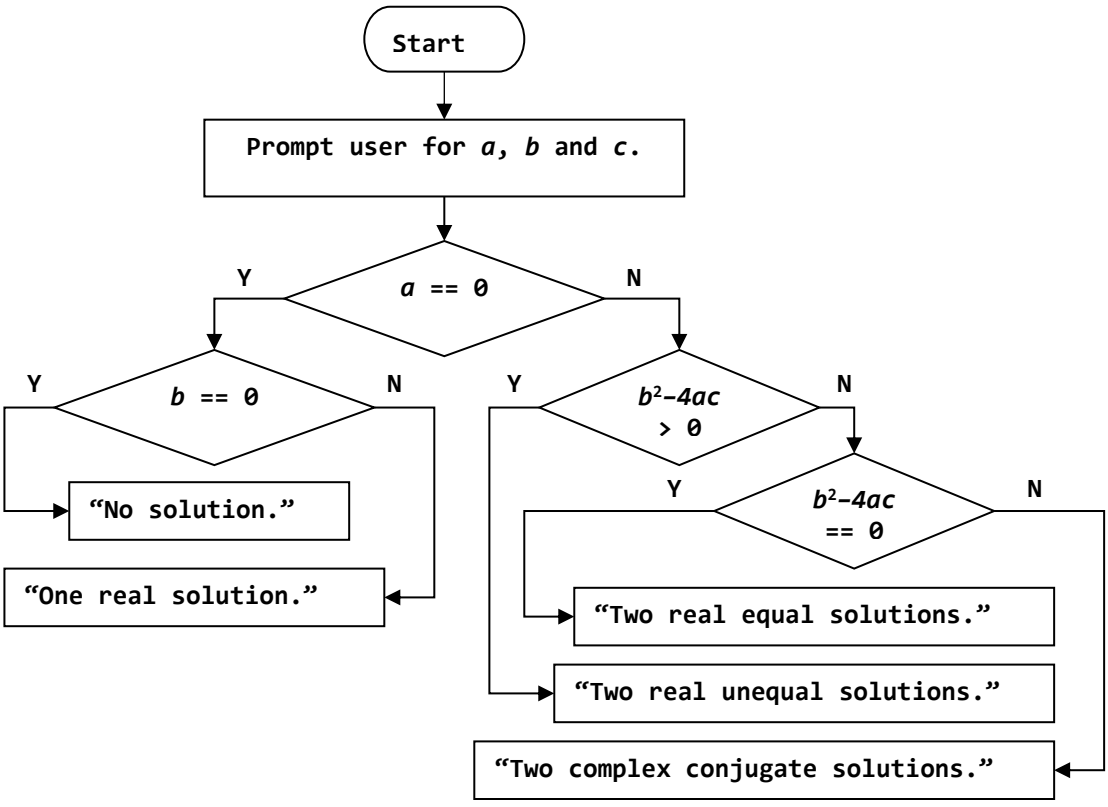
$$ax^2 + bx + c = 0$$

with real-numbered values of the constant coefficients *a*, *b* and *c* entered by the user. Your application should rely on the standard quadratic formulas:

$$x_1 = \frac{-b + \sqrt{b^2 - 4ac}}{2a}$$

$$x_2 = \frac{-b - \sqrt{b^2 - 4ac}}{2a}$$

Structure the program as shown in the following Flow Chart. It is recommended that you perfect the structure of the program before including the calculations.



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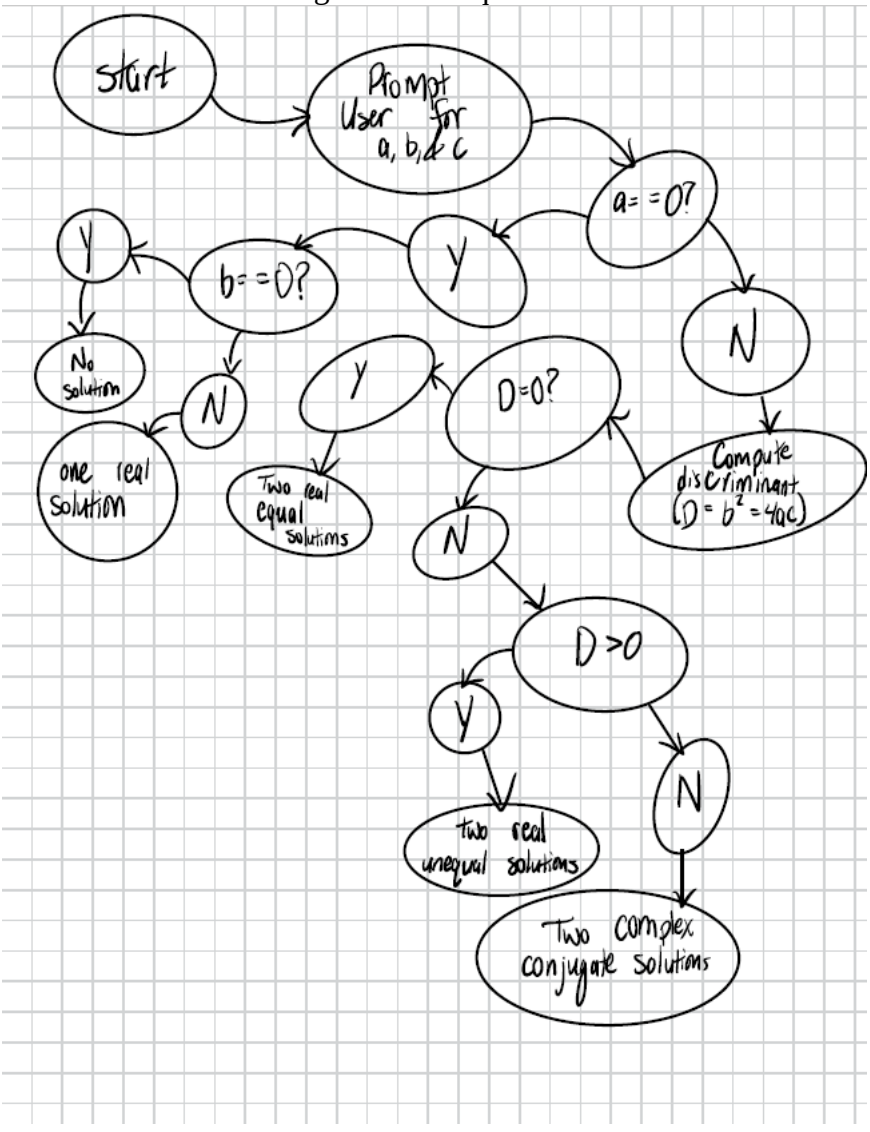
Following a common practice among electrical engineers, use the symbol *j* in your display to represent the square root of –1.

Document the program with the following sections:

A. A brief written description of the purpose of the program including the inputs, the outputs and a description of the structure of the programming.

The Quadratic Equation Solver calculates solutions for equations of the form ax^(2) +bx +c = 0. The user enters values for a, b, and c, while the program determines the types of solutions based on the discriminant (b^(2) -4ac). It then displays two real solutions, one repeated real solution, or two complex solutions. If a = 0, then the program checks if the equation is linear or has no solution. The program uses simple conditions to classify and display the correct result.

B. A flowchart including all of the equations for the various solutions.



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C. Source Code:

```
#include <iostream>
#include <cmath>

using namespace std;
int main()
{
    double a, b, c;

    // User input
    cout << "Quadratic Equation Solver" << endl;
    cout << "Enter coefficient a: ";
    cin >> a;
    cout << "Enter coefficient b: ";
    cin >> b;
    cout << "Enter coefficient c: ";
    cin >> c;

    // Check if it's a valid quadratic equation
    if (a == 0) {
        if (b == 0) {
            cout << "No solution." << endl;
        }
        else {
            double x = -c / b;
            cout << "One real solution: x = " << x << endl;
        }
        return 0;
    }

    // Compute the discriminant
    double discriminant = b * b - 4 * a * c;

    // Determine the nature of the roots
    if (discriminant > 0) {
        double x1 = (-b + sqrt(discriminant)) / (2 * a);
        double x2 = (-b - sqrt(discriminant)) / (2 * a);
        cout << "Two real unequal solutions: x1 = " << x1 << ", x2 = " << x2
<< endl;
    }
    else if (discriminant == 0) {
        double x = -b / (2 * a);
        cout << "Two real equal solutions: x = " << x << endl;
    }
    else {
        double realPart = -b / (2 * a);
        double imaginaryPart = sqrt(-discriminant) / (2 * a);
        cout << "Two complex conjugate solutions: x1 = " << realPart << " + "
<< imaginaryPart << "j, ";
        cout << "x2 = " << realPart << " - " << imaginaryPart << "j" << endl;
    }

    system("pause");
    return 0;
}
```

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D. Screen capture images showing the operation of the program for each of the possible solutions.

